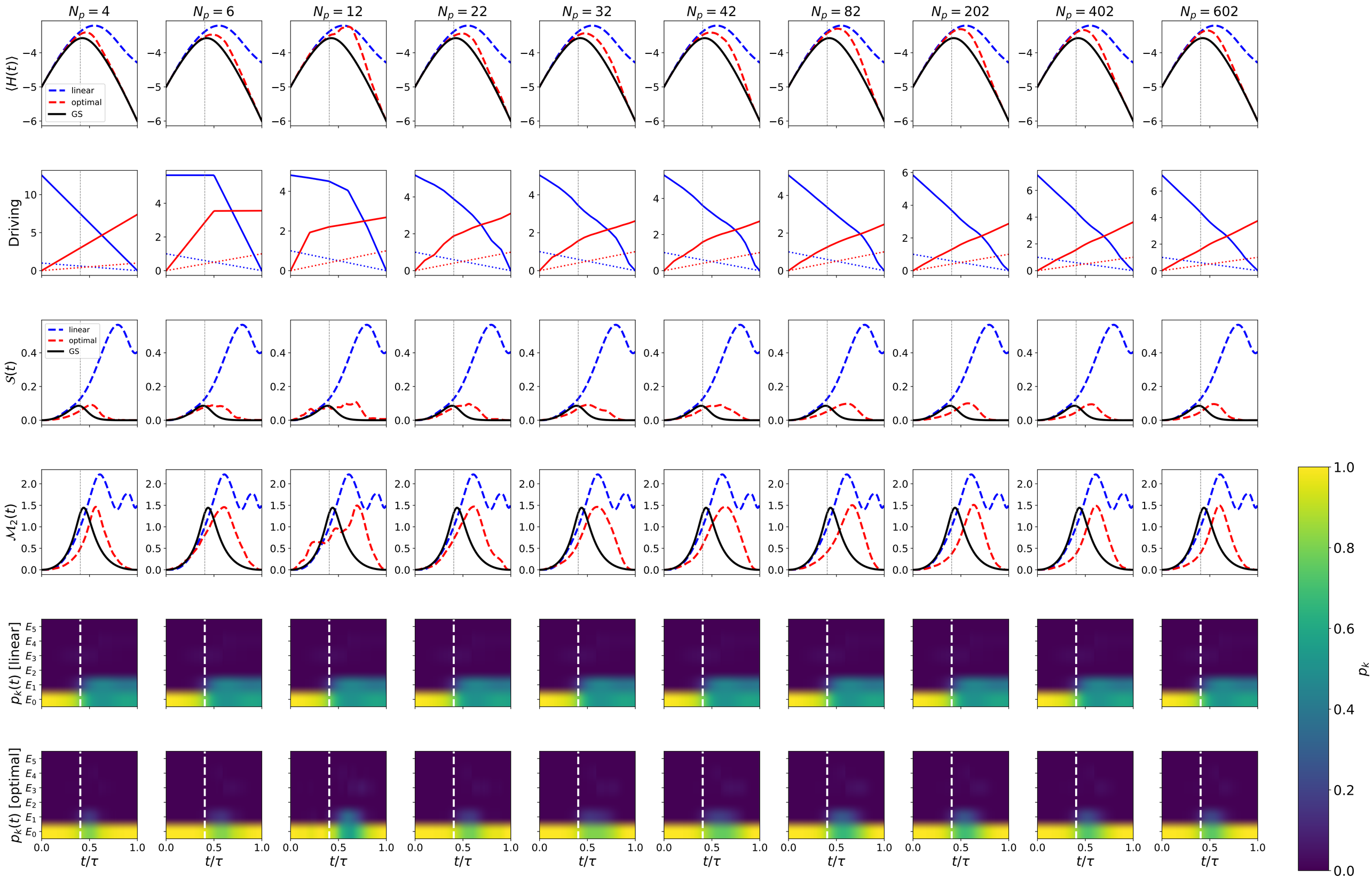Energy, drivings, $\mathcal{S}$, $\mathcal{M}_2$ \& populations vs N_p $(\tau = 5.0)$